

th-ass_1

1) What is List? How will you reverse a list?

Ans)

- A **list** is a data structure used to store multiple items in a single variable.
- It can hold **different types of data** (numbers, strings, etc.)
- List is ordered.
- There several ways to reverse list.
- slicing
- Or reverse function[reverse()]

th-ass_2

2)How will you remove last object from a list?

ans)

- There are various ways to remove last element from a list.
- `pop()` → removes last item and returns it (most used)
- `del list[-1]` → deletes last item
- `[:-1]` → creates a new list without last item.

th-ass_3

3) Suppose list1 is [2, 33, 222, 14, and 25], what is list1 [-1]?

ans)

- In Python, negative indexing starts from the end of the list
- So, [-1] represents 25..

th-ass_4

4) Differentiate between append () and extend () methods?

ans)

- Append ~
 - Adds one element.
 - it takes input Any object.
 - It adds single item at a command.
 - where it allows nested list.
- Extend ~
 - Adds multiple elements.
 - it only takes iterable objects as input.
 - it adds multiple items at a command.
 - it flattens the added elements.

th-ass_5

6) how will you compare two lists?

ans)

- The sign of `[==]` is used to compare two list.
- which checks whether both lists have the **same elements in the same order**.
- Where `/=` checks that the lists aren't equal.

th-ass_6

19) what is tuple? Difference between list and tuple.

ans)

- A **tuple** is a built-in data type in Python used to store multiple items in a single variable.
- A tuple is an **ordered and unchangeable collection** of elements.

DIFFERENCE BETWEEN TUPLE AND LIST

- **List** is mutable, while **Tuple** is immutable.
- **List** uses square brackets [], while **Tuple** uses round brackets ().
- **List** has more built-in methods, while **Tuple** has fewer methods.
- **List** is used when data may change, while **Tuple** is used when data should not change.

th-ass_7

32) How will you create a dictionary using a tuple.

ans)

- A dictionary can be created using a tuple of key-value pairs with the `dict()` function.
- Each tuple inside must contain two elements: the first is the key and the second is the value.

th-ass_8

36) How do you traverse through a dictionary object in python?

ans)

- A dictionary in Python can be traversed using a **for** loop.
- You can iterate through **keys** and **values**, or **both keys and values**.

th-ass_9

37) How do you check the presence of a key in a dictionary?

ans)

- By the function `get()` we can check the presence of a key in a dictionary.
- With use of this operator you can check the presence of key or else it would return none.

th-ass_10

44) Why do you use the Zip () method in python?

ans)

- The `zip()` method in Python is used to **combine two or more iterables element by element**.
- It pairs the first elements together, second elements together, and so on.
- It is used for **parallel iteration** and **creating pairs or dictionaries from lists**.

th-ass_11

53) How do you perform pattern matching in python?

ans)

- In Python, **pattern matching** is used to compare data structures against patterns and execute code based on the match.
- From **Python 3.10 onwards**, this is done using the **match statement**.
- Different patterns can be made by alphabetical or numerical designs in python.

th-ass_12

54) What is lambda function in python? What we call a function which is incomplete version of a function?

ans)

- The one line or small functions are called lambda functions.
- The incomplete version of python function is empty function or also called to be pass.

th-ass_13

55) How Many Basic Types Of Functions Are Available In Python?

ans)

- There are two types of functions Built-in and Userdefine.
 - User define functions are the functions made by the user for his purpose.
 - Built-in functions are provided pre-built in the python language.

th-ass_14

56) How can you pick a random item from a list or tuple?

ans)

- By using the built-in library called random we can pick a random item in python.
- Taking different modules from random library we can get random item from list or tuple.

th-ass_15

57) How can you pick a random item from a range?

ans)

- We can use the random function and give it in the range to get random item in python.
- Also by a module of random function called randint.
- By this method we can generate random item from specific range.

th-ass_16

58) How can you get a random number in python?

ans)

- We can generate a random item in python using a built-in library called random.
- With its function called random and randint we can generate random values.
- Also using uniform function in a specific range we can generate a random item.

th-ass_17

59) How will you set the starting value in generating random numbers?

ans)

- In the function of randint from random library we can get random item from its range.
- In this function we can give round brackets and with comma we can set the range.
- While creating manually we can generate random item by making list of the item values, And using the random function in that range.

th-ass_18

60) How will you randomizes the items of a list in place?

ans)

- To randomise the list, a random shuffle function can be used.
- Which shuffles the items in the list by changing its index.